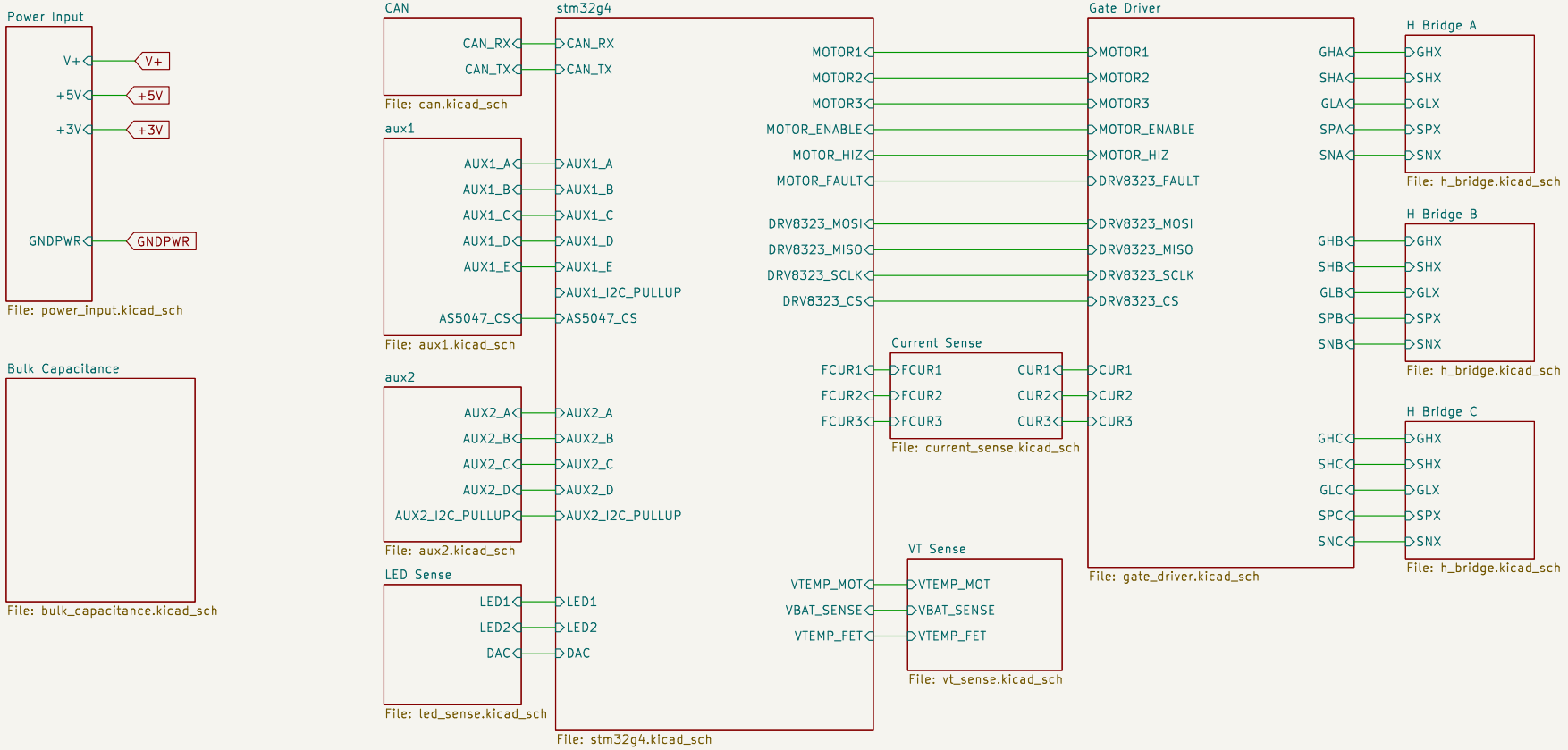
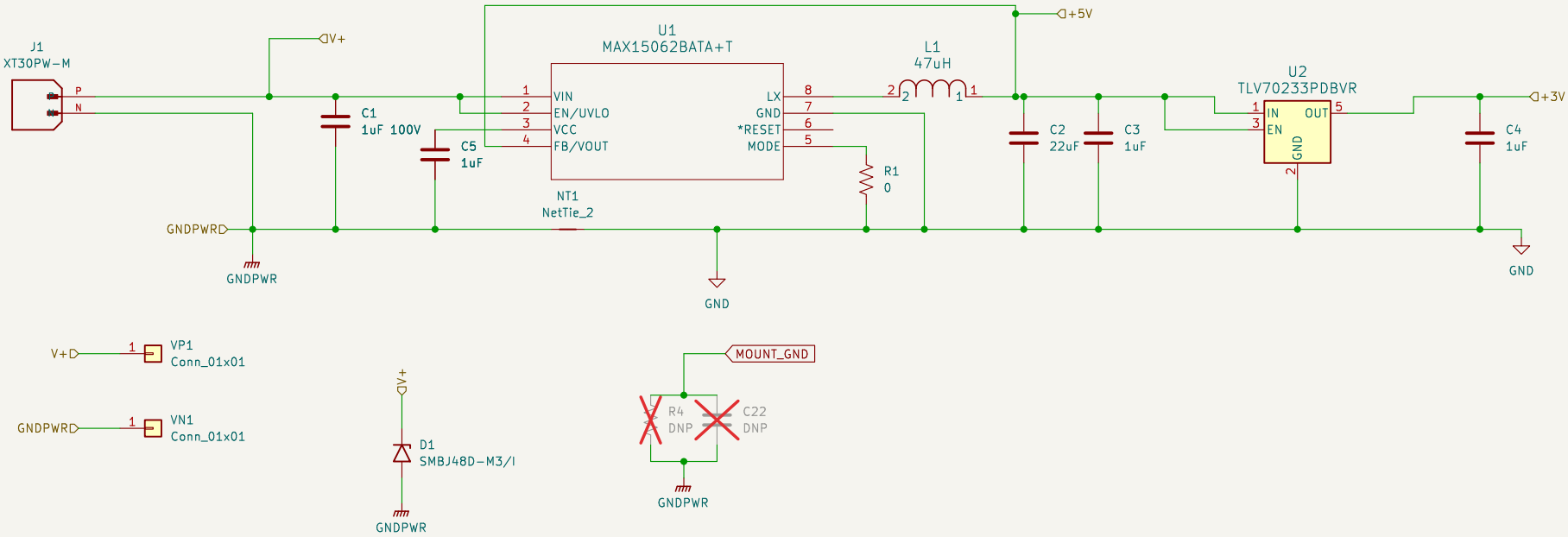


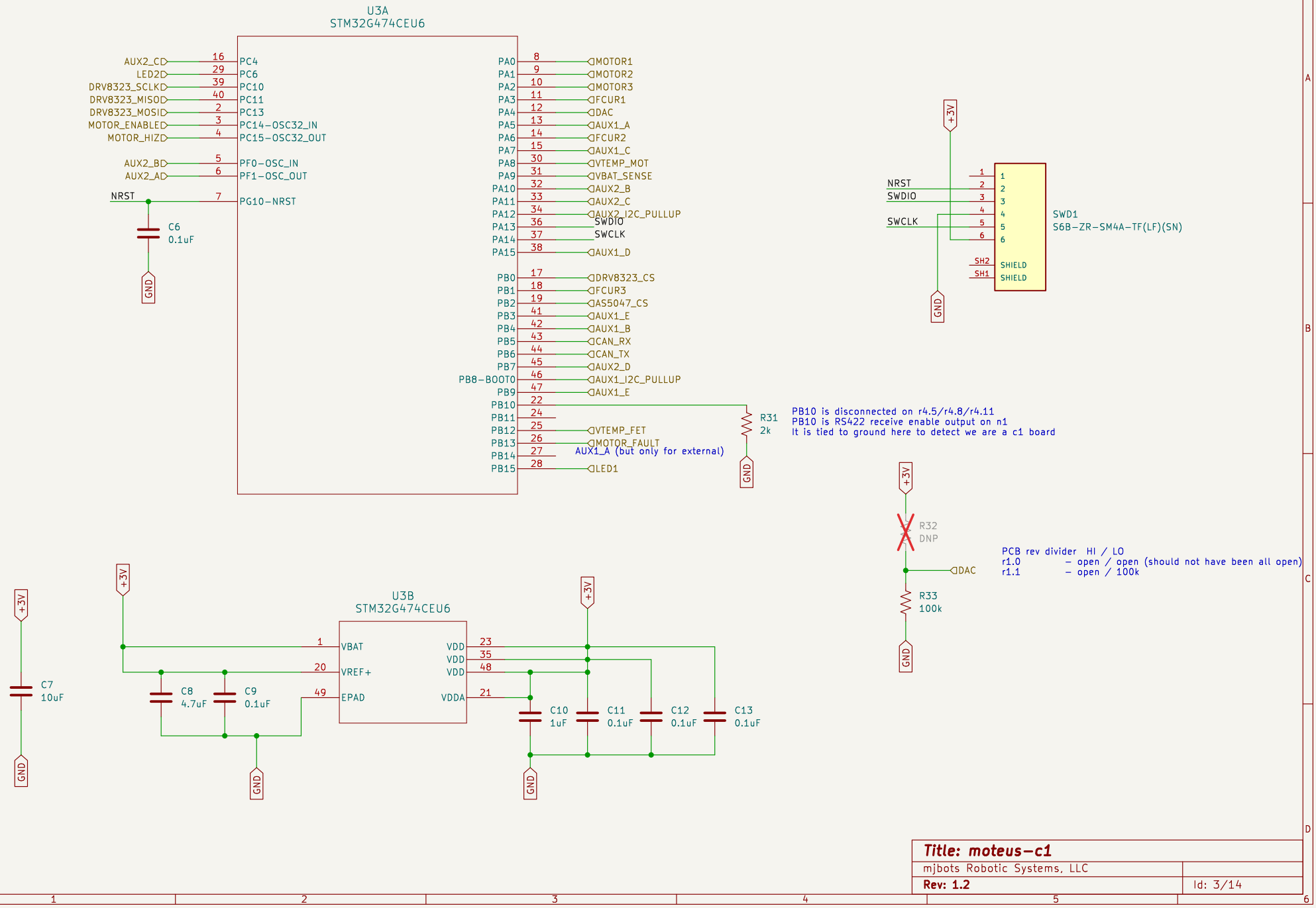
moteus-c1

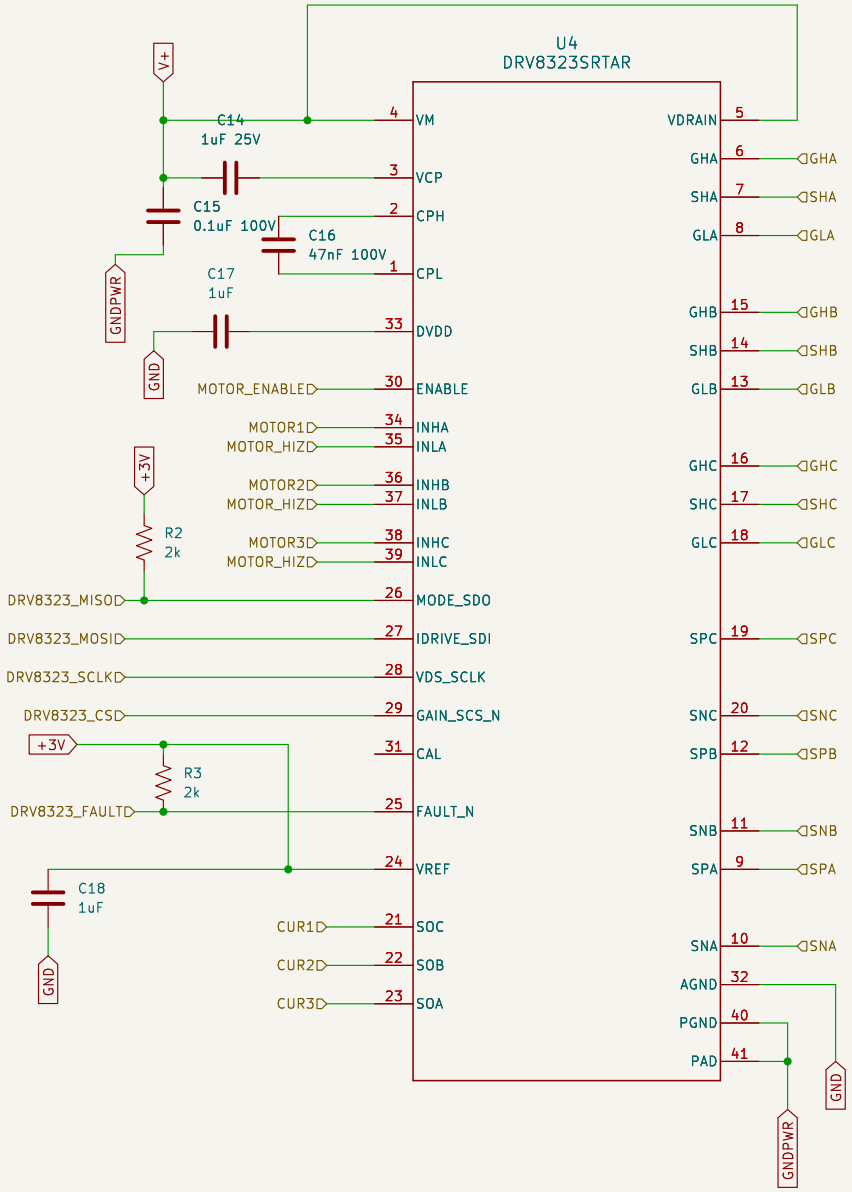
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Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2 + encoder DRAFT 0.2	Id: 1/14

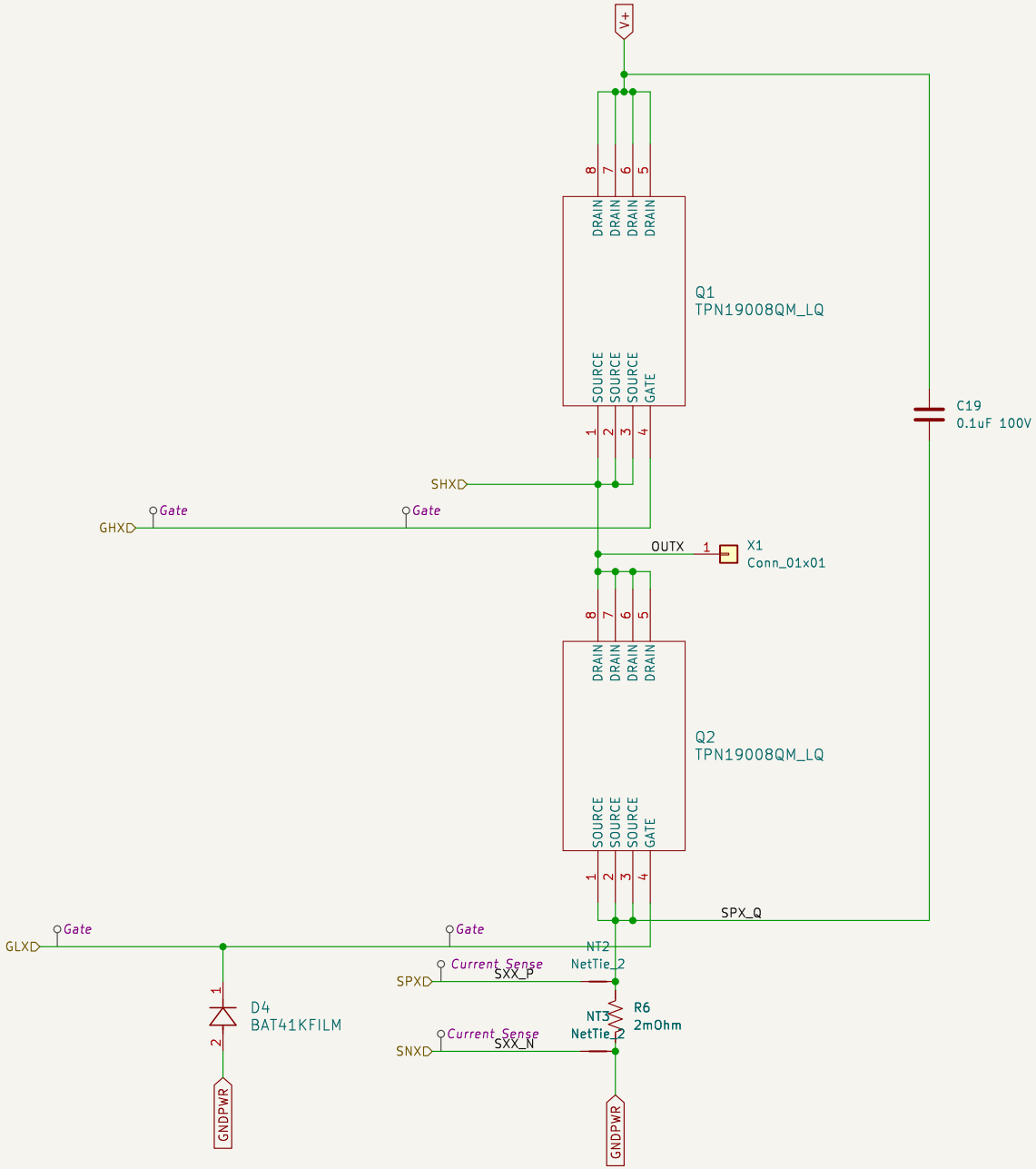


Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 2/14

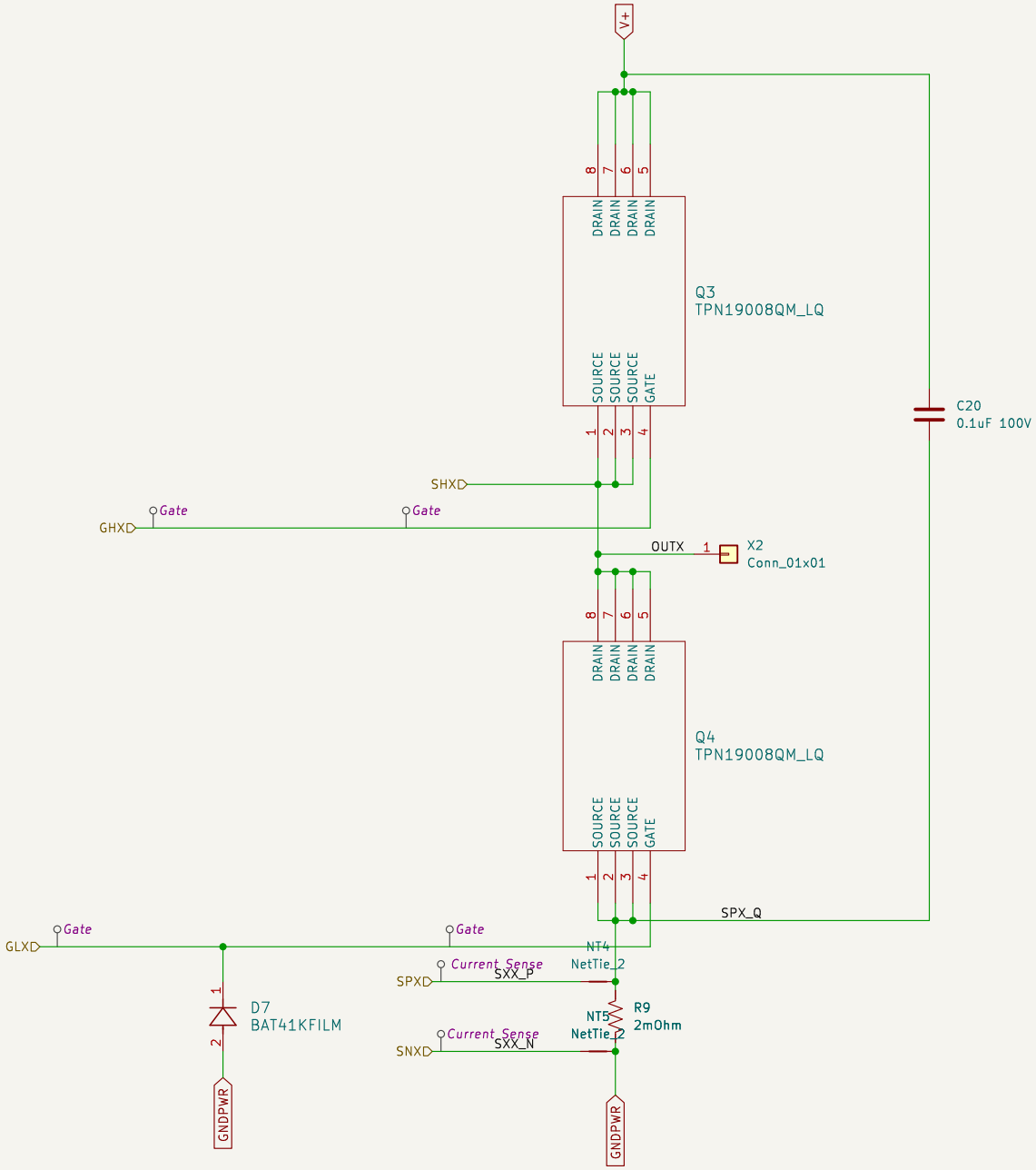

U3A
STM32G474CEU6
AUX2_C
LED2
DRV8323_SCLK
DRV8323_MISO
DRV8323_MOSI
MOTOR_ENABLE
MOTOR_HIZ
AUX2_B
AUX2_A
NRST
PC4
PC6
PC10
PC11
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
PG10-NRST
C6
0.1uF
GND
PA0 MOTOR1
PA1 MOTOR2
PA2 MOTOR3
PA3 FCUR1
PA4 DAC
PA5 AUX1_A
PA6 FCUR2
PA7 AUX1_C
PA8 VTEMP_MOT
PA9 VBAT_SENSE
PA10 AUX2_B
PA11 AUX2_C
PA12 AUX2_I2C_PULLUP
PA13 SWDIO
PA14 SWCLK
PA15 AUX1_D
PB0 DRV8323_CS
PB1 FCUR3
PB2 AS5047_CS
PB3 AUX1_E
PB4 AUX1_B
PB5 CAN_RX
PB6 CAN_TX
PB7 AUX2_D
PB8-BOOT0 AUX1_I2C_PULLUP
PB9 AUX1_E
PB10
PB11
PB12 VTEMP_FET
PB13 MOTOR_FAULT
PB14 AUX1_A (but only for external)
PB15 LED1
R31
2k
PB10 is disconnected on r4.5/r4.8/r4.11
PB10 is RS422 receive enable output on n1
It is tied to ground here to detect we are a c1 board
+3V
NRST
SWDIO
SWCLK
SWD1
S6B-ZR-SM4A-TF(LF)(SN)
SHIELD
R32
DNP
DAC
R33
100k
PCB rev divider HI / LO
r1.0 – open / open (should not have been all open)
r1.1 – open / 100k
U3B
STM32G474CEU6
VBAT
VREF+
EPAD
VDD
VDDA
C7
10uF
C8
4.7uF
C9
0.1uF
C10
1uF
C11
0.1uF
C12
0.1uF
C13
0.1uF
Title: moteus-c1
mjbots Robotic Systems, LLC
Rev: 1.2
Id: 3/14



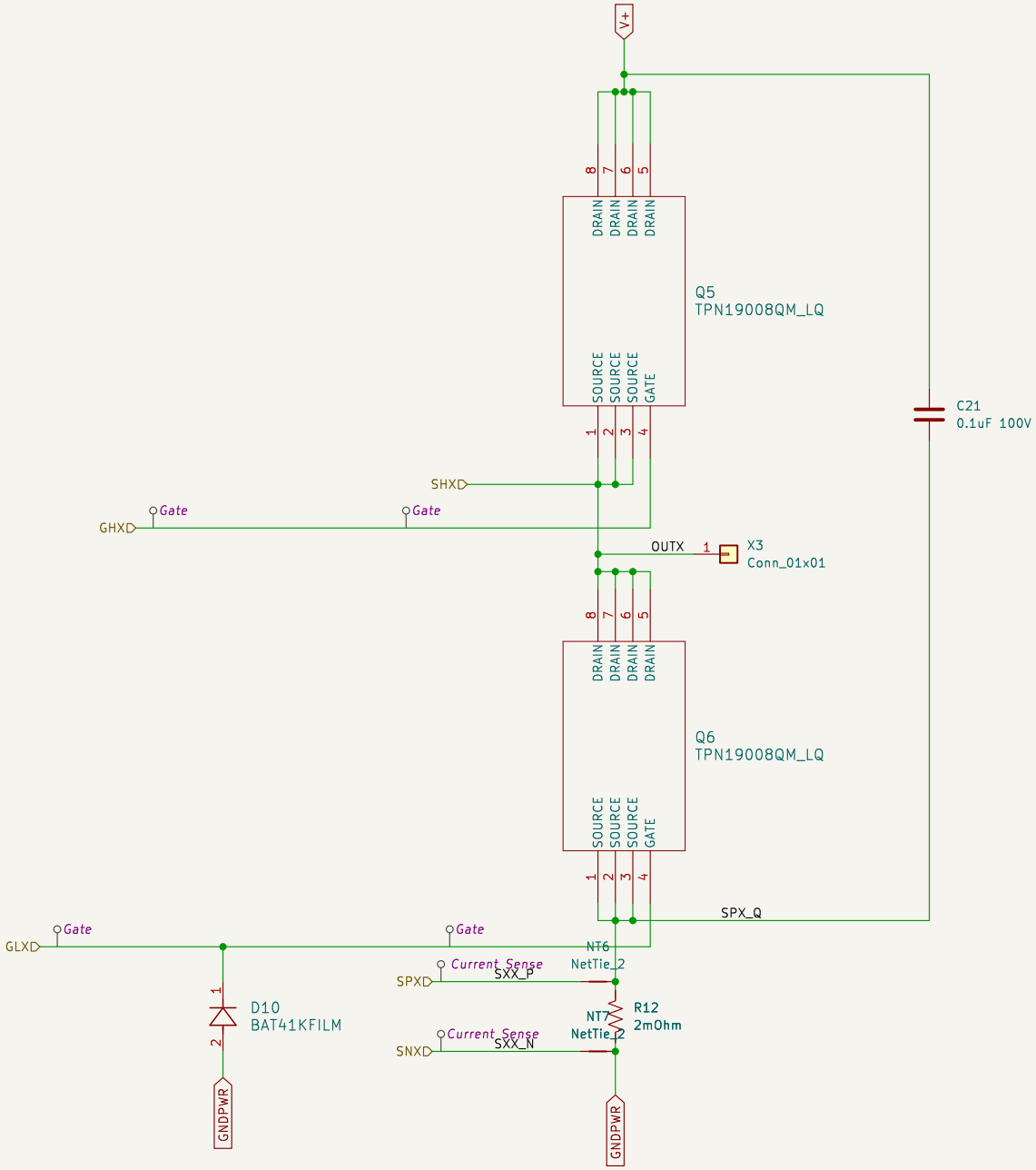
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mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 4/14



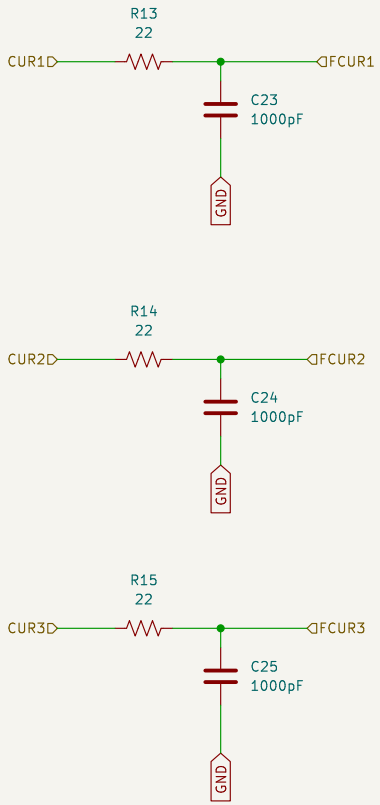
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mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 5/14



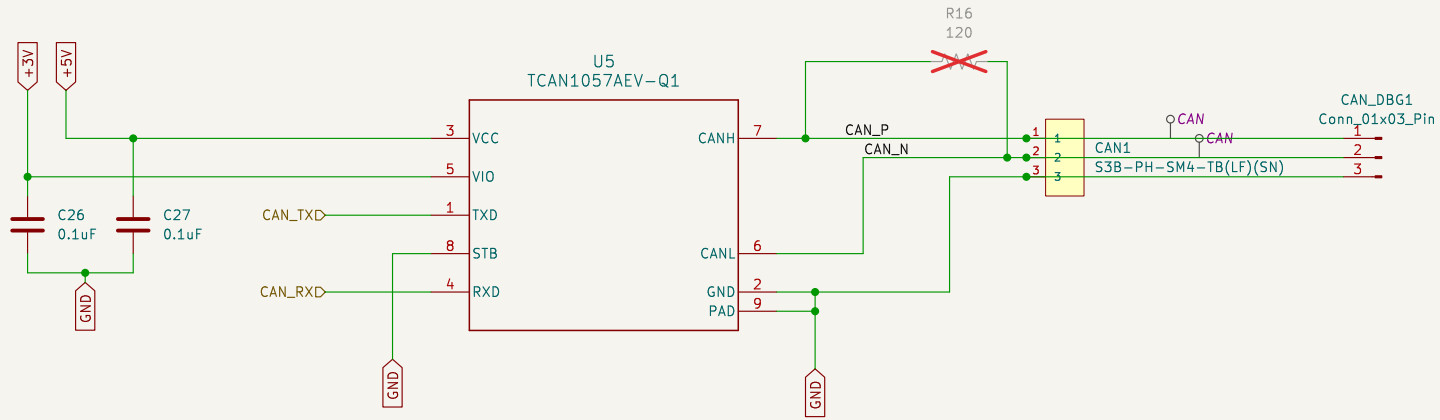
Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 6/14



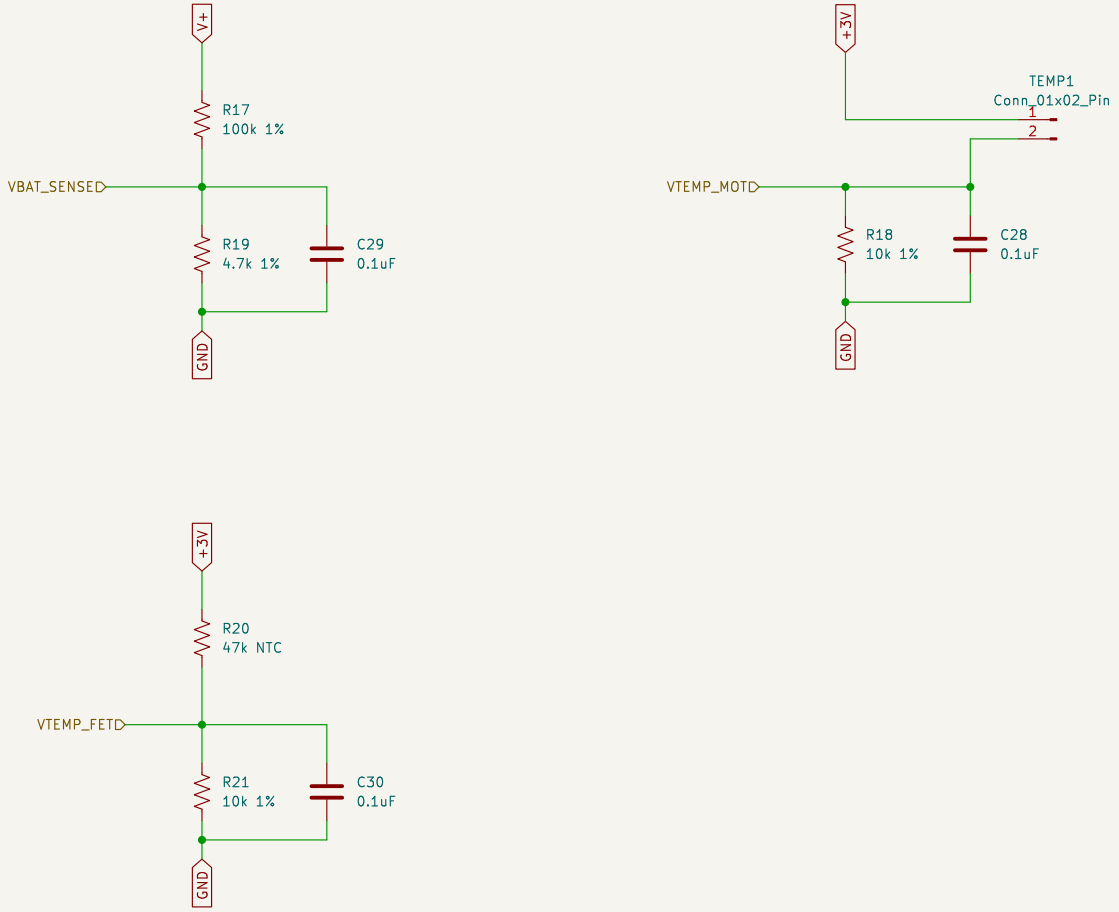
Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 7/14



Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 8/14



Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 9/14



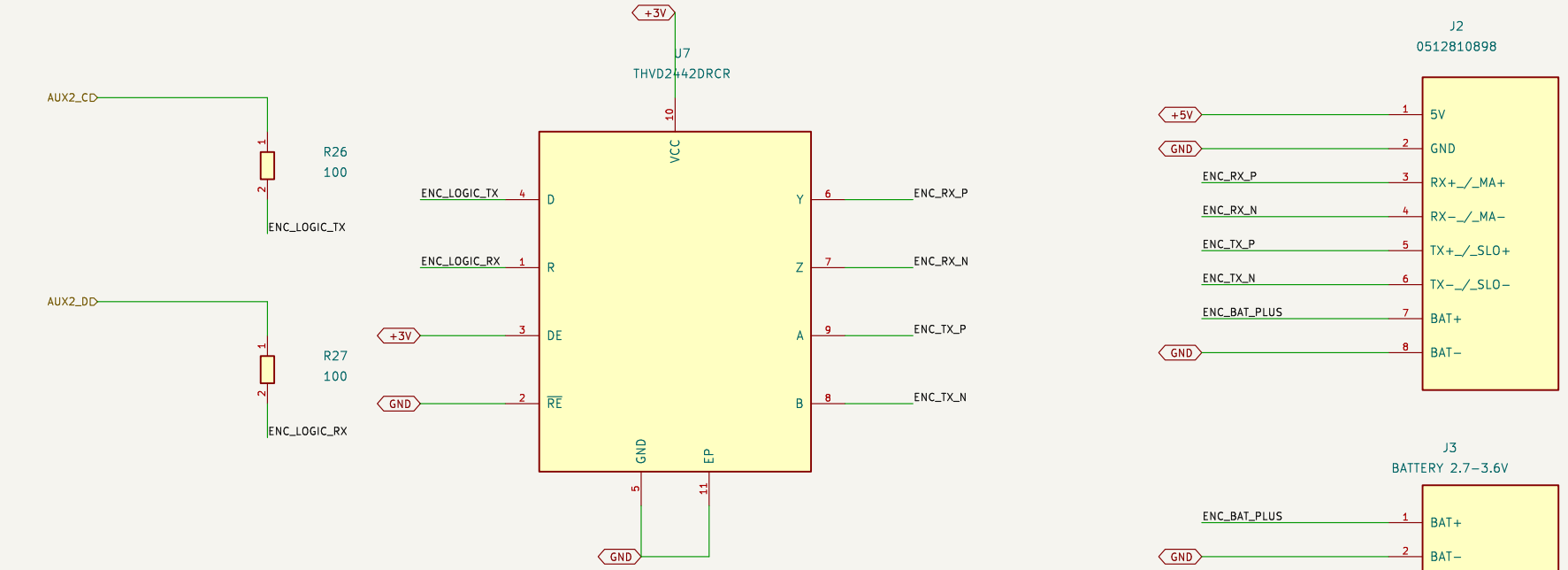
Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 10/14

ARCHIBALD | INTEGRATED C1 ENCODER INTERFACE

REVIEW DRAFT 0.2 — NOT FOR FABRICATION

Replaces stock AUX2 connectors; processor connections pass through retained R26 / R27.

C: PC4 UART TX / PA11 BiSS clock
D: PB7 UART RX / BiSS data



Unused AUX2 functions retired:

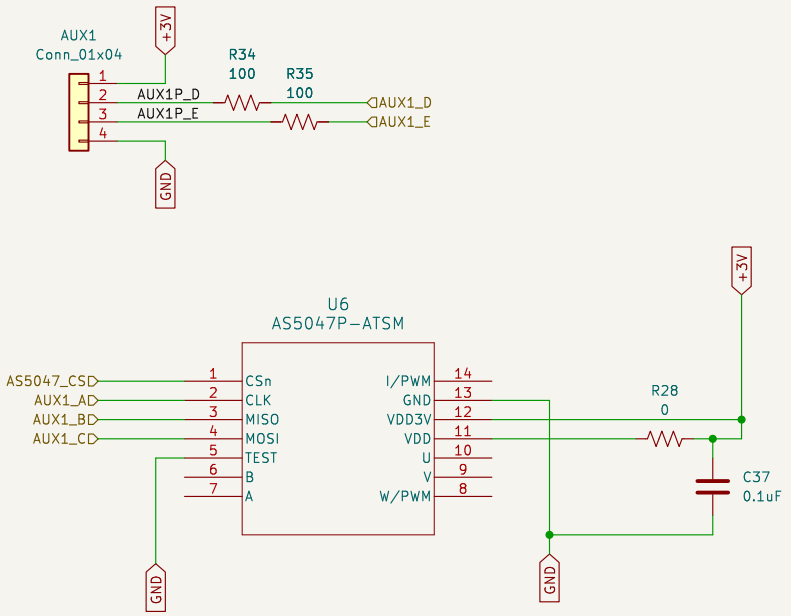
AUX2_AD
AUX2_BD
AUX2_I2C_PULLUP

3.3 V rail is named +3V in C1.
DE high / RE low; EP mapped to pad 11.
U7 footprint and FPC contact orientation: review.

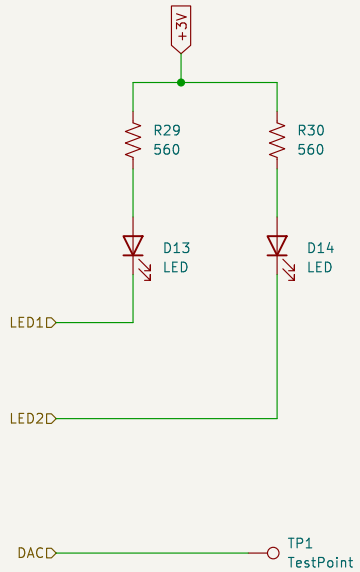
R36: receive termination, DO NOT POPULATE.
No added transmit termination or bias network.
J3: connector versus solder-pad choice pending.
Battery positive isolated from C1 supply rails.

Reference: Stephen Bell, Flux Riser Board Design, 2026-10-03. Power/load and signal-integrity review still required.
C50 replaces Flux C3; U7 replaces Flux U1; J2 replaces Flux J1; R36 replaces Flux R1. PCB integration not performed.

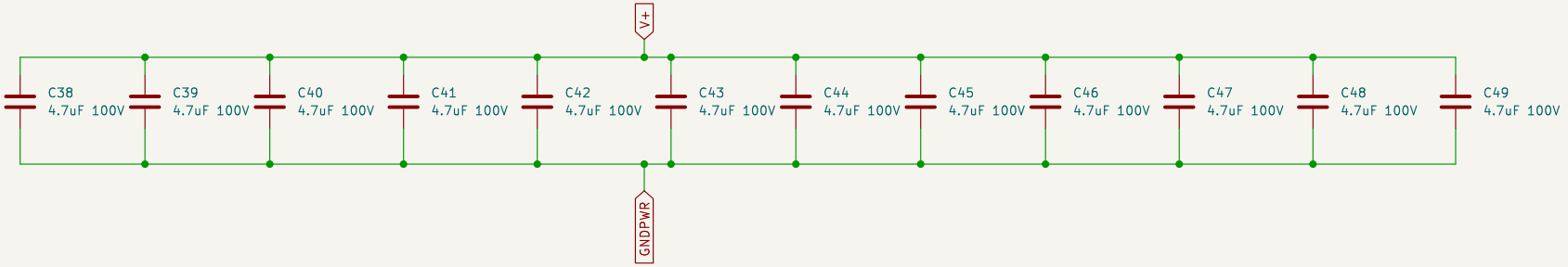
Title: C1 encoder integration — review only	
mjbots Robotic Systems, LLC	
Rev: DRAFT 0.2	Id: 11/14



Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 12/14



Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 13/14



Title: moteus-c1	
mjbots Robotic Systems, LLC	
Rev: 1.2	Id: 14/14